DAY 29 | PRE-READ

How Good Is Our AI, Really?

A light read before Day 29. About 15 minutes, no code required.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Why read this?

Every customer will ask: "How do we know the AI is right?" Tomorrow you learn to answer with numbers instead of impressions, and to find the specific ways a system fails, including inventing facts and treating some users worse than others.

1. The tasting panel

A restaurant chain does not judge a new recipe by one chef tasting it once. A panel tastes many plates against a checklist: flavour, temperature, presentation. AI evaluation works the same way: a fixed set of realistic questions, a checklist of qualities, and scores you can compare over time.

2. Two places things go wrong

Stage	What can fail	Everyday version
Finding information	The right passage is never found	The librarian brings the wrong book
Writing the answer	The answer ignores or twists the passage	The student misquotes the right book

Measuring the two stages separately tells you what to fix.

3. Faithfulness

A faithful answer only says things the sources support. If an assistant adds "and employees get a free gym membership" when no document mentions it, that is a **hallucination**, even if it sounds plausible. We measure faithfulness by checking each claim against the sources.

4. AI checking AI

Reading thousands of answers by hand is slow, so teams often use one AI model to grade another, with a clear marking scheme. Useful, but these judges have habits, such as preferring longer answers, so we check them against human grades on a sample.

5. Fair for everyone

An assistant can be accurate on average and still perform badly for certain groups, for example questions written in Japanese. Breaking results down by group reveals this.

Key idea: measure, then read the failures. Numbers tell you where to look; reading tells you why.

6. Think about these before class

- Write five questions a new employee would ask an HR assistant, with the correct answers.
- How would you know if an AI answer was made up?
- Which user groups in your organisation might be served worse by an AI system?

Hakaru (測る): to measure